Topic 3 – Processes, techniques and tools

3.1a Heat treatments and alloying

Spec: Heat treatments – hardening and tempering, case hardening, annealing, normalising (including use of specialist tools); Alloying (including use of specialist tools)

Key facts

• **Steps to harden and temper a chisel blade?** Heat tip to cherry red → quench in water/oil (hard, brittle) → clean bright with emery → reheat gently behind the tip until the edge turns the correct oxide colour (dark straw ~240 °C) → quench.
• **Which steels can be hardened?** Medium and high carbon steels (over about 0.4% carbon). Mild steel cannot – it must be case hardened.
• **Tempering colour for cutting tools such as chisels?** Pale to dark straw (≈230–250 °C) for very hard edges; purple/blue (≈280–300 °C) for tough springs and screwdrivers.
• **Case hardening process in school?** Heat mild steel to cherry red → dip in carbon powder (Casenit) → reheat/re-dip 2–3 times → reheat → quench in cold water.
• **Purpose of annealing?** Make metal soft and ductile, relieve work hardening so it can be bent/beaten further.
• **How do you anneal copper vs steel?** Copper: heat to dull red and quench or air cool. Steel: heat to cherry red and cool very slowly in the hearth/ashes.
• **Purpose of normalising?** Refines grain structure, removes internal stresses (e.g. after forging or welding), giving uniform properties.
• **Name two specialist tools/equipment for heat treatment.** Brazing hearth with gas torch, furnace/muffle kiln, blacksmith's tongs, quenching tank, pyrometer, leather apron, face shield, heat-resistant gloves.
• **Name the two heat treatments that make work-hardened steel tougher and less brittle.** Annealing and tempering (also normalising). Not quenching or hardening.
• **Why alloy metals?** To improve strength, hardness, corrosion resistance, machinability, colour or lower melting point.

Process	Suitable metals	Method	Result	Typical use
Hardening	Medium & high carbon steel	Heat to cherry red (≈ 760–900 °C, just above upper critical temp.), soak, **quench** in water/brine/oil	Very hard, wear resistant, **brittle**	Chisels, punches, drills, files
Tempering	Hardened carbon steel	Clean bright, reheat gently to 230–300 °C judged by oxide colours, then quench	Less brittle, **tougher**, slightly less hard	Same tools after hardening
Case hardening	Mild (low carbon) steel	Heat to cherry red, dip in carbon powder (or pack carburise in a box for hours), repeat, reheat and quench	Hard, wear-resistant **skin**; tough, ductile **core**	Gears, cams, axles, hinge pins, one-off school tools
Annealing	Steel, copper, brass, aluminium	Heat (steel: cherry red; copper: dull red; aluminium: until soap mark/blackened wood chars) then cool **very slowly** (steel in furnace/ashes)	Softest, most ductile state; removes work hardening	Before bending, beating, spinning or deep pressing
Normalising	Steel	Heat above upper critical temperature, cool in **still air**	Refined, uniform grain; internal stresses relieved	After forging, welding or heavy machining

Colour	≈ Temperature	Use
Pale straw	230 °C	Scribers, lathe tools, razors (hardest)
Dark straw	240–250 °C	Drills, punches, chisels
Brown	260 °C	Wood chisels, plane blades
Purple	270–280 °C	Knives, axes, cold chisels
Blue	300 °C	Springs, screwdrivers (toughest)

Key terms

Hardening	Heating medium/high carbon steel to cherry red (about 760–900 °C, just above the upper critical temperature) and quenching rapidly in water or oil. Makes it very hard but brittle.
Tempering	Reheating hardened steel to 230–300 °C (judged by oxide colours, pale straw to blue) then quenching. Reduces brittleness, increases toughness, slightly reduces hardness.
Case hardening	Adding carbon to the surface of low carbon (mild) steel then quenching, giving a hard wear-resistant skin and a tough core.
Annealing	Heating metal and cooling it very slowly (in the furnace or ashes) to make it as soft and ductile as possible and remove work hardening.
Normalising	Heating steel above its upper critical temperature and cooling in still air. Refines the grain and relieves internal stresses; less soft than annealing.
Quenching	Cooling hot metal rapidly by plunging it into water, brine or oil.
Critical temperature	The temperature range (about 723–900 °C for steel) where its crystal structure changes.
Alloying	Melting and mixing a metal with other elements to create an alloy with improved properties.

Exam tip (2024 Q1a): "Name two heat treatments that restore toughness to work-hardened steel" – answer **annealing, tempering, normalising**. *Hardening* and *quenching* scored zero.

3.1b Printing processes

Spec: Printing – offset lithography, flexography, screen-printing, gravure (including use of specialist tools)

Key facts

• **Principle of offset lithography?** Oil and water don't mix: the image area attracts oil-based ink and repels water; ink is offset from plate to rubber blanket to paper.
• **Name the three cylinders in an offset litho press.** Plate cylinder, blanket cylinder, impression cylinder.
• **Uses of offset lithography?** Books, magazines, newspapers, leaflets, posters, packaging – medium to long runs of high quality.
• **Flexography – order of rollers?** Ink pan/fountain roller → anilox roller (+ doctor blade) → plate cylinder → impression cylinder.
• **Uses of flexography?** Packaging: corrugated boxes, food wrappers, plastic film, labels, carrier bags, newspapers.
• **Advantages of flexography over gravure?** Cheaper photopolymer plates, quicker set-up, economical for shorter runs, prints on uneven/varied surfaces.
• **Why is gravure used for long runs?** Engraved copper/chrome cylinders are very durable and give consistent, high-quality images over millions of copies.
• **Disadvantage of gravure?** Extremely high cost and lead time for engraving cylinders – only viable for very long runs.
• **Benefits of screen printing?** Prints on almost any surface (fabric, wood, glass, plastic, curved), low set-up cost, thick vivid ink, good for short runs.
• **Drawbacks of screen printing?** One screen per colour, drying time between colours, slow and labour intensive, fine detail difficult.
• **Two uses of a densitometer?** Measure ink density/thickness and colour saturation; calibrate presses and check print quality consistency.

	Offset lithography	Flexography	Gravure	Screen printing
Principle	Planographic: oil and water repel	Relief: raised image on flexible plate	Intaglio: image engraved as recessed cells	Stencil: ink pushed through mesh

	Offset lithography	Flexography	Gravure	Screen printing
Image carrier	Thin aluminium plate (made by CTP – computer-to-plate)	Photopolymer rubber plate	Copper-plated, chrome-faced steel cylinder	Polyester mesh screen + photo-emulsion stencil
Run length	Medium–long (≈1 000–1 000 000)	Medium–long, flexible	Very long (millions)	Short / one-off / batch
Quality	High, sharp	Good (improving)	Excellent, consistent	Good, thick vibrant ink, limited fine detail
Substrates	Paper and board (flat, smooth)	Packaging, plastic film, corrugated board, foil, labels	Paper, film, foil	Almost anything: fabric, wood, glass, plastic, metal, curved
Set-up cost	Medium	Low–medium	Very high	Low
Typical products	Books, magazines, leaflets, posters	Food packaging, carrier bags, labels, corrugated boxes	Glossy magazines, stamps, banknotes, wallpaper, decor laminates	T-shirts, posters, signs, PCBs, printed glassware

Key terms

Offset lithography	Printing based on oil and water not mixing. The image on a plate attracts oily ink, is transferred (offset) to a rubber blanket and then onto paper.
Flexography	Relief printing with a flexible photopolymer plate wrapped round a cylinder, inked by an anilox roller. Fast; prints on packaging, plastic film and corrugated board.
Gravure	Intaglio printing – the image is engraved as tiny cells into a copper-plated cylinder. Ink sits in the cells; very high quality and very long runs.
Screen printing	Ink is forced with a squeegee through a fine mesh screen; a stencil blocks areas that should not print.
CMYK	Cyan, magenta, yellow and key (black) – the four process colours used to print full-colour images.
Registration marks	Cross-hair targets used to line up each colour plate accurately.
Densitometer	Instrument that measures the optical density of printed ink to check ink thickness, colour saturation and print consistency, and to calibrate presses.
Anilox roller	Engraved roller in flexography that meters a precise, even film of ink onto the plate.
Doctor blade	A flexible blade that scrapes excess ink off a roller or gravure cylinder.

3.1c Casting

Spec: Casting – sand (to include investment), die, resin, plaster of Paris (including use of specialist tools)

Key facts

• **Steps to prepare a sand mould (split pattern)?** Half pattern in drag + parting powder → ram sand, strike off → turn over, add cope, second half pattern, sprue and riser pins → ram sand → separate, remove pattern and pins → cut gates/runners → reassemble.

• **Why use green sand?** Sand + clay + water holds its shape when rammed, is cheap and reusable, and is permeable so gases escape.

• **Why are patterns made slightly oversized?** Metal shrinks as it cools (shrinkage allowance) and some surfaces need machining allowance.

• **Advantages of investment casting?** Extremely accurate and detailed, smooth finish, complex shapes, little machining – jewellery, turbine blades, golf club heads.

• **Advantages of die casting?** High volume, precise, good surface finish, thin walls, reusable die; but very expensive dies, only non-ferrous metals (low melting point).

• **Why can't steel be die cast economically?** Its very high melting point would damage the steel die.

• **What is resin casting used for?** Small batches of prototypes, models, jewellery, encapsulation – polyester/polyurethane/epoxy resin poured into silicone moulds.

• **What is plaster of Paris casting used for?** Making moulds, sculptures, architectural mouldings, prototype models and medical casts.

• **Name two control measures when pouring aluminium in school.** Pour over a sand tray/floor; use crucible tongs and pouring shank; screens/exclusion zone; local fume extraction; preheat tools to remove moisture.

• **Specialist tools for sand casting?** Moulding flask (cope and drag), rammer, strike-off bar, sprue pins, vent wire, trowel, crucible, crucible tongs and shank, furnace.

Process	Mould	Materials	Volume	Pros	Cons
Sand casting	Expendable sand mould around a reusable pattern	Aluminium, cast iron, bronze, steel	One-off to batch	Cheap tooling, large parts, any metal	Rough surface, less accurate, needs fettling/machining
Investment (lost wax)	Expendable ceramic shell around a wax pattern	Steel, precious metals, superalloys	Batch–mass	Very accurate, fine detail, complex shapes, smooth	Slow, costly per part, pattern lost each time
Die casting	Permanent steel die (pressure or gravity)	Aluminium, zinc, magnesium (non-ferrous)	Mass	Precise, thin walls, good finish, fast	Very expensive dies, not for high-melting metals
Resin casting	Silicone rubber or other flexible mould	Polyester, polyurethane, epoxy resins	One-off–batch	Fine detail, cheap moulds, room temperature	Fumes (COSHH), brittle resins, slow curing
Plaster of Paris	Rubber, plastic or clay mould	Plaster (calcium sulfate + water)	One-off	Cheap, safe, detailed, used to make moulds	Fragile, heavy, absorbs moisture

Key terms

Casting	Pouring a liquid material (molten metal, resin, plaster) into a mould where it solidifies into the shape of the cavity.
Pattern	A replica of the product (often jelutong or MDF) used to make the cavity in a sand mould. Slightly oversized to allow for shrinkage, with draft angles.
Cope and drag	The top (cope) and bottom (drag) boxes of a sand-casting moulding flask.
Sprue and riser	The sprue (runner) is the channel molten metal is poured into; the riser lets air escape and feeds metal as the casting shrinks.
Investment casting	Lost-wax casting: a wax pattern is coated in ceramic slurry, the wax is melted out, and metal is poured into the ceramic shell. Very accurate, fine detail.
Die casting	Molten non-ferrous metal (aluminium, zinc) forced under pressure (or by gravity) into a reusable steel mould (die). High volume, precise.
Parting powder	French chalk dusted on patterns and between sand layers so they separate cleanly.
Flash	Thin excess material that seeps out at the join line of a mould and must be trimmed.

3.1d Machining, stamping and pressing

Spec: Machining – milling/routing, drilling, turning, stamping, pressing (including use of specialist tools)

Key facts

• **Name four lathe operations.** Facing off, parallel turning, taper turning, drilling/boring, knurling, parting off, thread cutting.

• **Key safety points for a centre lathe?** Remove chuck key, guards down, goggles, tie hair/remove loose clothing, never measure while rotating, clear swarf with a brush not hands, know the emergency stop.

• **Milling vs routing?** Milling: slower, rigid machines, metals; routing: high speed, lighter, for wood/MDF/plastics. Both use rotating cutters.

• **How is a car body panel made?** Steel coil straightened → blanked → automated press forms it between dies (pressing) → piercing/trimming → ejected.

• **Preparing square timber for wood turning?** Mark centres with diagonals, saw kerfs for the drive centre, plane to octagon, mount between centres, set tool rest, rough with gouge, smooth with skew chisel.

• **Wood lathe tools?** Roughing-out gouge, spindle gouge, skew chisel, parting tool, scraper; drive (fork) centre and live/revolving tail centre.
• **What is a centre punch used for before drilling?** Makes a small dent so the drill tip starts in the correct place without wandering.
• **Benefits of press forming?** Very fast, consistent, accurate, suitable for mass production, little waste; but tooling is very expensive.

Key terms

Milling	Removing material with a rotating multi-tooth cutter while the work is fed past it, to make flat faces, slots, steps and pockets in metal or plastic.
Routing	Similar to milling but at high speed, mainly for wood, MDF and plastics – profiles, grooves, rebates and edges.
Turning	Shaping a rotating workpiece on a lathe with a single-point cutting tool (metal) or hand-held chisels/gouges (wood) to create cylindrical forms.
Drilling	Cutting a round hole with a rotating twist drill or other bit (pillar drill, hand drill, CNC).
Stamping/blanking	Cutting or punching shapes out of sheet metal with a punch and die in a press.
Pressing	Forming sheet metal (often after blanking) into a 3D shape between matched male and female dies under high pressure.
Facing off	Lathe operation cutting the end of a bar flat and square.
Feed and speed	Feed rate = how fast the cutter moves through the work; spindle speed = rpm. Both depend on material and cutter diameter.

3.1e Moulding processes

Spec: Moulding – blow moulding, injection moulding, vacuum forming, extrusion, rotational moulding (including use of specialist tools)

Key facts

• **Injection moulding steps?** Granules in hopper → screw carries them past heaters → melt → screw/ram injects into mould → held under pressure, cooled → mould opens, ejector pins remove part.
• **Products made by injection moulding?** Phone cases, bottle caps, LEGO, chair shells, games controller casings, buckets, toys.
• **Drawbacks of injection moulding?** Very expensive moulds and long lead times, only economical for mass production, design changes need new moulds.
• **Blow moulding steps?** Parison/preform placed in open split mould → mould closes → compressed air inflates plastic against cold walls → cools → mould opens, bottle ejected and trimmed.
• **Products made by blow moulding?** PET drinks bottles, HDPE milk bottles, shampoo bottles, fuel cans.
• **Vacuum forming steps?** Mould on platen (lowered) → clamp sheet → heat until it sags → raise platen → vacuum pulls sheet over mould → cool → remove and trim.
• **Mould features for vacuum forming?** Draft angles, rounded corners, no undercuts, vent holes in recesses, smooth rigid surface.
• **Disadvantages of vacuum forming?** Walls thin at corners/deep recesses, lots of waste sheet, limited detail, open-backed shapes only, needs trimming.
• **Products made by extrusion?** Pipes, guttering, window frames, curtain track, cable insulation, plastic rod, aluminium profiles.
• **Benefits of rotational moulding?** Large hollow seamless parts, even wall thickness, stress-free, cheap moulds, inserts possible – kayaks, water tanks, traffic cones, toy cars.

Process	Product shape	Volume	Tooling cost	Typical products
Injection moulding	Solid, complex 3D, fine detail	Mass	Very high (steel moulds)	Casings, caps, toys, chair shells
Blow moulding	Hollow, thin-walled, narrow neck	Mass	High	Bottles, containers, fuel tanks
Vacuum forming	Thin open shells, no undercuts	One-off–batch (school) to mass (industrial)	Low (MDF/wood moulds) to medium	Trays, blister packs, organisers, bath liners
Extrusion	Continuous constant cross-section	Continuous/ mass	Medium (die)	Pipes, window frames, rod, cable sheath
Rotational moulding	Large, hollow, seamless	Batch–mass	Low–medium (aluminium moulds)	Kayaks, tanks, cones, playground slides

Key terms

Injection moulding	Thermoplastic granules are melted by heaters and a rotating screw, then injected under high pressure into a closed, cooled steel mould.
Blow moulding	A hot tube of plastic (parison) or a preform is clamped in a split mould and inflated with compressed air to form a hollow product.
Vacuum forming	A heated thermoplastic sheet is drawn over a mould by vacuum to make thin shell shapes.
Extrusion	Molten polymer (or hot metal) is forced continuously through a shaped die to make long products of constant cross-section.
Rotational moulding	A measured charge of polymer powder is heated in a mould rotating about two axes so it coats the inside, making large hollow seamless products.
Parison	The hollow tube of softened plastic used in extrusion blow moulding.
Draft angle	A slight taper (about 3–5°) on mould sides so the product can be removed easily.
Ejector pins	Pins that push a finished injection moulding out of the mould. They leave small circular marks.
Sprue	The channel (and waste plastic) through which molten plastic enters the mould.

3.1f Lamination and marking out

Spec: Lamination (including use of specialist tools); Marking out techniques – woods, metals, polymers, paper and boards (including use of specialist tools)

Key facts

• **Steps to laminate a curved hardwood rocker (one-off)?** Make former → release agent/cling film → cut veneers (soak/steam if needed) → apply glue (PVA/UF/epoxy) → stack in former → clamp evenly from centre outward or bag press → wipe excess glue → cure → release and trim.
• **Why laminate rather than cut a curve from solid wood?** Strong curve with grain following the shape (no short grain), less waste, stable, can use thin decorative veneers.
• **Which adhesives are used for wood laminating?** PVA (interior), urea formaldehyde/Cascamite (gap filling, water resistant), epoxy resin.
• **Marking out tools for wood?** Pencil, try square, marking gauge, mortise gauge, cutting gauge, marking knife, sliding bevel, steel rule, tape.
• **Marking out tools for metal?** Scriber, engineer's square, odd-leg callipers, dividers, centre punch, dot punch, surface plate, scribing block, engineer's blue.
• **Marking out on polymers?** Leave protective film on; draw with a spirit-based marker or chinagraph pencil, or scribe lightly; use templates.
• **Marking out on paper and board?** Pencil, steel rule, set squares, compass, templates, cutting mat, craft knife; creasing bar/scoring for folds.
• **Why use a template or jig when marking out batches?** Speeds up marking out, ensures every part is identical, reduces errors and skill needed.

Material	Lines	Tools
Wood	Pencil, then knife lines for cuts (prevents splintering)	Steel rule, tape, try square, sliding bevel, marking gauge, mortise gauge, cutting gauge, marking knife, dovetail template, combination square
Metal	Scribed lines, made visible with engineer's blue	Scriber, engineer's square, steel rule, odd-leg (jenny) callipers, dividers, centre punch, dot punch, surface plate and scribing block/height gauge, vee blocks
Polymers	Marker or chinagraph pencil on the protective film; light scribing	Steel rule, try/engineer's square, templates, spirit marker, scriber, dividers
Paper & board	Light pencil, fold lines scored	Steel rule, set squares, compass, protractor, templates, cutting mat, safety rule, craft knife, scoring/creasing tool

Key terms

Lamination (wood)	Gluing thin layers (veneers/laminates) together over a former so they keep a curved shape when the adhesive cures.
Former	A mould or jig (often MDF) the shape of the finished curve; can be male/female pair or a single form.
Bag press	Vacuum bag that applies even atmospheric pressure to laminations over a former.
Datum	A reference edge, line or point from which all measurements are taken to avoid accumulated errors.
Scriber	Hardened steel point used to scratch fine lines on metal and plastics.
Odd-leg callipers	Callipers with one stepped leg used to scribe lines parallel to an edge on metal.
Engineer's blue	Marking fluid painted on metal so scribed lines show clearly.

Exam tip: Marks are awarded for stages **in the correct order** – a maximum of 5 out of 6 if the order is wrong. For a bespoke product describe one-off methods, not mass production.

3.2 Specialist measuring tools

Spec: Marking, cutting and mortise gauges; Odd leg, internal and external callipers; Squares (set, try, engineers and mitre); Micrometer and vernier callipers; Densitometer; Dividers; Jigs and fixtures; Go and no-go gauges

Key facts

• **How do you read a vernier calliper?** Read the main scale mark just before the vernier zero, then find the vernier line that aligns exactly with a main scale line and add it (e.g. 23 mm + 0.36 = 23.36 mm).
• **How do you read a metric micrometer?** Sleeve: whole mm + any half mm visible; thimble: line aligned with datum × 0.01 mm. Total = sleeve + half + thimble (e.g. 7 + 0.5 + 0.22 = 7.72 mm).
• **Function of the ratchet on a micrometer?** Applies the same light measuring force each time so readings are consistent and the tool isn't over-tightened.
• **Go and no-go gauge – how used?** GO end must enter/fit; NO-GO end must not – the part is within tolerance. Fast, no skill, no reading needed – ideal for quality control in mass production.
• **Jig vs fixture?** Jig holds work and guides the tool; fixture only holds/locates the work.
• **Uses of a mortise gauge?** Marks two parallel lines at once for the width of a mortise and tenon, with a single pin on the other side for marking gauge use.
• **Cutting gauge vs marking gauge?** Cutting gauge has a small blade that cuts across the grain cleanly (e.g. for dovetail baselines); marking gauge has a pin for along the grain.
• **Internal vs external callipers?** Internal (legs turn out) measure bores/insides; external (legs bow in) measure outside diameters – both transfer sizes to a rule.
• **Which squares are on the spec?** Set square (drawing 30/60/45°), try square (wood), engineer's square (metal, all steel, more accurate), mitre square (45°).
• **What are dividers used for?** Scribing circles and arcs on metal/plastic, stepping off equal distances, transferring measurements.
• **What does a densitometer measure?** Optical density of ink on a print, used to check and control colour quality and calibrate presses.

Tool	Description	Material	Precision / use
Marking gauge	Stock slides on a stem with a steel pin	Wood	Lines parallel to an edge, along the grain
Cutting gauge	Like a marking gauge but with a small knife blade	Wood	Cuts clean lines across the grain (dovetail/tenon shoulders)
Mortise gauge	Two adjustable pins (plus one on reverse)	Wood	Both sides of a mortise/tenon in one stroke
Odd-leg callipers	One straight point, one stepped leg	Metal, plastic	Scribe lines parallel to an edge; find centres
Internal callipers	Legs turned outwards	Any	Transfer internal diameters to a rule
External callipers	Legs bowed inwards	Any	Transfer outside diameters (e.g. turning)
Set square	45° or 30/60° triangle	Drawing	Drawing angles and verticals
Try square	Steel blade in wooden/plastic stock, 90°	Wood	Marking/checking right angles
Engineer's square	All-steel, precision ground	Metal	Accurate 90° marking and checking
Mitre square	Blade fixed at 45° (135°)	Wood	Marking/checking mitre joints
Vernier callipers	Main + vernier scale (or digital)	All	0.02 mm; external, internal and depth
Micrometer	Anvil, spindle, sleeve, thimble, ratchet	All	0.01 mm; very accurate external sizes
Densitometer	Optical instrument	Printing	Ink density, colour saturation, press calibration
Dividers	Two pointed steel legs	Metal, plastic, drawing	Scribe circles/arcs, step off equal spaces
Jigs and fixtures	Custom holding/guiding devices	All	Repeatability in batch production
Go/no-go gauges	Fixed plug or gap gauges with two limits	All	Fast pass/fail tolerance checks in QC

Key terms

Accuracy	How close a measurement or product is to the true/specified value.
Precision	How consistently repeatable measurements are, and the fineness of the scale (e.g. 0.01 mm).
Tolerance	The permitted variation in a dimension, e.g. 25 ± 0.05 mm. Upper and lower limits.
Vernier callipers	Measuring tool with main and sliding vernier scales, reading to 0.02 mm (or 0.05 mm); measures external, internal and depth dimensions.
Micrometer	Screw-thread measuring tool reading to 0.01 mm (0.001 mm on vernier micrometers) for very accurate external dimensions.
Go/no-go gauge	Fixed gauge for quick pass/fail checking of a dimension: the GO end must fit, the NO-GO end must not.
Jig	A device that holds the work AND guides the tool (e.g. a drilling jig) so repeated parts are identical.
Fixture	A device that holds and locates the work securely in the correct position while it is machined or assembled (does not guide the tool).

3.3 Drawing and communicating design

Spec: Pictorial drawing methods for representing 3D forms – isometric, 2-point perspective; Working drawings for communicating 2D technical information – 3rd angle orthographic projection, triangulation; Nets (developments) for communicating information about 3D forms in a 2D format; Translation between working drawings, pictorial drawings and nets (developments); Report writing

Key facts

• **Third angle: where are the plan and end elevations?** Plan directly above the front elevation; right-hand view to the right of the front (left view to the left).
• **Draw the third angle projection symbol (describe it).** A truncated cone: its side view (a trapezium tapering to the right) on the LEFT and its end view (two concentric circles) on the RIGHT.
• **Line types in BS 8888?** Thick continuous: visible outlines. Thin continuous: dimension, projection, hatching lines. Thin dashed: hidden detail. Thin chain (long dash–dot): centre lines.
• **Rules for dimensioning?** Dimensions in mm (no unit on each), numbers above the line read from the bottom/right, thin dimension lines with filled arrows, small gap from object, no duplication, overall dimensions outside smaller ones, Ø for diameter, R for radius.
• **Isometric angles?** Verticals vertical; horizontal edges at 30° left and right. Measure true lengths along the axes. Circles become ellipses.
• **Why can a 45° roof not be drawn at 45° in isometric?** Angles are distorted in isometric – plot the end points using coordinates along the axes and join them.
• **How to draw 2-point perspective?** Horizon line with two vanishing points → draw the nearest vertical edge → project top and bottom to both VPs → mark widths and draw verticals → join back to opposite VPs.
• **Net line conventions?** Solid lines = cut; dashed (or chain) lines = fold/crease; glue tabs angled ~45°; apertures shown as

cut-outs.
• **What should a technical report include?** Title, aim/purpose, introduction/context, method, results (tables, graphs), analysis, conclusion, recommendations, references; formal, objective, third person.
• **Exploded drawing purpose?** Shows how parts assemble in their correct order and relationship, often for instructions.

Line	Use
Thick continuous	Visible outlines and edges
Thin continuous	Dimension lines, projection/extension lines, leader lines, hatching
Thin dashed	Hidden detail
Thin chain (long dash, dot)	Centre lines, lines of symmetry
Thick chain	Cutting planes for sections

Key terms

Isometric projection	Pictorial drawing where vertical lines stay vertical and horizontal edges are drawn at 30° to the horizontal. All three axes are at 120° and true lengths are measured along them.
2-point perspective	Pictorial drawing with two vanishing points on a horizon line; vertical lines stay vertical and other edges converge – the most realistic view.
Orthographic projection	A set of 2D views (front elevation, plan, end elevation) drawn at right angles, giving true sizes for manufacture.
Third angle projection	Orthographic layout used in the UK/USA where the plan is drawn ABOVE the front elevation and the right-hand end view to the RIGHT of it.
Net / development	A flat 2D pattern that folds into a 3D form – showing cut lines, fold lines, glue tabs and apertures.
Triangulation	A development method for complex surfaces (e.g. transition pieces from square to round) by dividing the surface into triangles and finding the true length of each edge.
BS 8888	British Standard for technical product documentation – line types, dimensioning, projection symbols, title blocks.
Crating	Drawing a box (crate) the overall size of an object first, then cutting away/adding detail.

3.4a Adhesives

Spec: Adhesives – contact adhesive, acrylic cement, epoxy resin, polyvinyl acetate (PVA), hot melt glue, cyanoacrylate (superglue), polystyrene cement (including use of specialist tools)

Key facts

• **Best adhesive for wood joints indoors?** PVA – cheap, strong (stronger than the wood), non-toxic, water-based, easy clean up; needs cramping for ~1 hour, full cure 24 h; not waterproof (unless exterior grade).
• **Contact adhesive – how is it used?** Apply thin coat to BOTH surfaces, leave until touch dry (~10–15 min), then press together – instant bond, no repositioning. Used for laminates, veneers, leather, rubber, fabric.
• **Acrylic cement – how does it work?** Solvent (e.g. Tensol) softens acrylic surfaces which fuse; applied with a syringe/brush and drawn in by capillary action; very strong, clear joint.
• **Epoxy resin adhesive – properties?** Two-part (resin + hardener), very strong, bonds dissimilar materials (metal, wood, glass, ceramics), gap filling, waterproof, heat resistant; expensive, slow cure, measure accurately.
• **Hot melt glue – benefits?** Sets in seconds, no cramping, fills gaps, bonds dissimilar materials, can be automated, cheap, clear; but weaker, can be stringy, burn risk, softens in heat.
• **Cyanoacrylate – properties and uses?** Superglue: bonds in seconds with moisture, very strong on small areas, bonds skin; brittle, poor gap filling – small repairs, model making, medical skin glue.
• **Polystyrene cement – uses?** Solvent cement for polystyrene (HIPS) – plastic model kits, vacuum formed parts; VOCs so use ventilation.
• **Why must surfaces be prepared before gluing?** Clean, dry, grease-free and keyed (roughened) surfaces give the adhesive more area to grip – stronger bond.
• **Specialist tools for adhesives?** Glue guns, applicator bottles/brushes, syringes (acrylic cement), mixing sticks & spatulas (epoxy), G-cramps, sash cramps, band cramps, spreaders, rollers (contact adhesive).

Adhesive	Joins	Properties	Application & tools	Health & safety
PVA (polyvinyl acetate)	Wood, paper, card, fabric	Water-based, cheap, strong (often stronger than wood), dries clear; slow (cramp 30–60 min, full cure 24 h); interior grade not waterproof	Bottle/brush; sash cramps, G-cramps; wipe excess with damp cloth	Low hazard; avoid eyes
Contact adhesive (e.g. Evo-Stik)	Laminates, veneers, rubber, leather, fabric, foam	Instant bond on contact, flexible, waterproof; no repositioning	Spread thinly on **both** surfaces, leave touch dry, then press/roll	Solvent-based – VOCs, highly flammable, ventilation (COSHH)
Acrylic cement (e.g. Tensol)	Acrylic to acrylic	Solvent weld, very strong, clear joint	Fine applicator/syringe along tight-fitting joint – capillary action; hold still	Toxic, flammable vapour; ventilation, gloves, goggles

Adhesive	Joins	Properties	Application & tools	Health & safety
Epoxy resin (e.g. Araldite)	Almost anything – metals, wood, glass, ceramics, polymers, dissimilar materials	Very strong, gap filling, waterproof, heat and chemical resistant	Mix equal parts resin + hardener on a card with a spatula; set 5 min–24 h	Skin irritant – gloves; ventilation
Hot melt glue	Wood, card, fabric, many polymers, dissimilar materials	Sets in seconds as it cools, fills gaps, no cramps, can be automated; weaker, softens if heated, stringy	Electric glue gun with glue sticks; industrial applicators on packaging lines	Burns from hot nozzle/glue; stand safely
Cyanoacrylate (superglue)	Small parts – metals, polymers, rubber, ceramics	Bonds in seconds (reacts with moisture), very strong in tension on small areas; brittle, poor gap filling	Single drop; press briefly	**Bonds skin and eyes instantly**; ventilation
Polystyrene cement	Polystyrene (HIPS)	Solvent weld	Tube or brush with fine applicator	VOCs, flammable – ventilation

Key terms

Adhesive	A substance that bonds surfaces together by adhesion (sticking to surfaces) and cohesion (internal strength).
Solvent cement	Adhesive that softens/dissolves the surfaces of polymers so they fuse together as the solvent evaporates (acrylic cement, polystyrene cement).
Curing	The chemical hardening of an adhesive (e.g. epoxy resin and hardener).
Open time	How long an adhesive remains workable after application.
Cramping time	How long parts must be held under pressure while the adhesive sets.
VOC	Volatile organic compounds – solvent vapours in some adhesives (contact adhesive, polystyrene cement) that are harmful to breathe – COSHH controls apply.
Capillary action	How thin acrylic cement is drawn into a tight joint between two pieces of acrylic.

Exam tip (2024 Q2): "Give two benefits of hot melt glue" – rapid set, no cramping, fills gaps, joins dissimilar materials, can be automated. Avoid vague "it's easy/cheap".

3.4b Mechanical fixings

Spec: Mechanical – screws, nuts, bolts, washers, rivets, press (including use of specialist tools)

Key facts

• **Name four screw head types.** Countersunk, round head, raised countersunk, pan head; drive types – slotted, Pozidriv/Phillips, hex socket, Torx.
• **Wood screw vs machine screw?** Wood screw: tapered, coarse thread, cuts into wood. Machine screw: parallel thread, fits a tapped hole or nut.
• **Why use a washer?** Spreads the load, protects the surface, prevents loosening (spring/shakeproof washers), seals (fibre/rubber).
• **Benefits of pop rivets?** Fitted from one side only, single quick operation, strong permanent joint, no finishing needed, join dissimilar sheets.
• **Snap head (solid) rivet process?** Drill holes, insert rivet, support head in a dolly/snap, peen the tail with a ball pein hammer, form with a rivet snap set.
• **Name nut types.** Hexagonal nut, wing nut (hand tightened), lock nut (nylon insert/nyloc), dome/cap nut, castle nut with split pin.
• **Joining a car wing to the body?** Spot welding, bolts/machine screws, self-tapping screws, rivets, adhesive bonding.
• **Tools for mechanical fixings?** Screwdrivers (correct tip), spanners/socket sets, Allen keys, taps and dies, pop rivet gun, hammer and dolly/rivet set, countersink bit, bradawl, torque wrench, arbor press.

Fixing	Type	Use	Notes
Wood screws	Temporary	Joining wood and manufactured boards, fixing fittings	Drill clearance + pilot hole; countersink for flush heads. Steel, stainless, brass (use brass/stainless in oak)
Machine screws	Temporary	Metal parts with tapped holes or nuts	Parallel thread (M3, M4, M6…); heads: pan, countersunk, cap/socket
Self-tapping screws	Semi-permanent	Sheet metal, plastics, car panels, electrical casings	Hardened, cut their own thread
Nuts and bolts	Temporary	Strong joints that can be dismantled	Hexagon head bolts; nuts include hex, wing (hand tightened), nyloc (locking), dome
Washers	–	Spread load, protect surface, lock (spring, shakeproof), seal	Used under nuts and screw heads
Rivets	Permanent	Sheet metal, leather, aircraft, bridges	**Snap head** (solid, formed with a hammer and rivet set), **pop/blind** (fitted from one side with a rivet gun)
Press / interference fit	Semi-permanent	Bearings, bushes, pins, gears on shafts	Part slightly larger than hole; pressed with an arbor/hydraulic press, or by heating/cooling one part

Key terms

Temporary fixing	Can be taken apart without damage (screws, nuts and bolts, KD fittings).
Permanent fixing	Cannot be separated without destroying it (rivets, welding, most adhesives).
Countersunk screw	Screw with a conical head that sits flush with the surface in a countersunk hole.
Self-tapping screw	Hardened screw that cuts its own thread in sheet metal or plastic.
Pilot hole	Small hole drilled for the screw thread so wood doesn't split; clearance hole is the size of the shank.
Pop (blind) rivet	Rivet fitted from one side only using a pop rivet gun – the mandrel is pulled and snaps, expanding the rivet.
Press fit (interference fit)	Joining parts by forcing a slightly oversized part into a smaller hole, held by friction.
Spring washer	Split washer that resists loosening from vibration.

Exam tip (2025 Q2a): Technical benefits score; "easy" and "cheap" do not.

3.4c Joining with heat

Spec: Heat – oxy-acetylene welding, MIG welding, brazing, hard soldering, soft soldering (including use of specialist tools)

Key facts

• **Welding vs brazing/soldering?** Welding melts the parent metals (fusion); brazing and soldering use a lower-melting filler and the parent metals do NOT melt.
• **Rank the processes by temperature.** Oxy-acetylene/MIG welding (melt steel ~1500 °C) > brazing (~850–900 °C) > hard (silver) soldering (~600–800 °C) > soft soldering (~180–250 °C).
• **Purpose of flux?** Cleans and stops oxides forming while heating so the filler wets and flows into the joint.
• **Steps for brazing mild steel?** Clean with emery → fit joint tightly, support on fire bricks → apply borax flux → heat joint (not the rod) to bright red → touch brass spelter to joint, flows in → cool → clean off flux residue.
• **MIG advantages?** Fast, continuous wire feed, easy to learn, clean welds without slag, can be automated/robotic (car bodies).
• **Oxy-acetylene hazards?** Very high temperature flame, flashback (fit flashback arrestors), explosive gases (store cylinders upright, chained), UV/glare – welding goggles, leather gloves, apron.
• **Why is soft solder used for electronics?** Low melting point won't damage components, good conductor, quick with a soldering iron; joints can be desoldered.
• **PPE for MIG welding?** Auto-darkening welding helmet (UV/IR), leather gauntlets, flame-resistant jacket/apron, boots; screens to protect others; fume extraction.
• **Specialist tools?** Oxy-acetylene torch, regulators, hoses & flashback arrestors; MIG set with wire feed and gas; brazing hearth and torch, fire bricks; soldering iron, solder sucker, helping hands; flux brush; tongs; welding helmet.

Process	Heat source	Filler	Approx. temp.	Parent metal melts?	Strength	Typical uses
Oxy-acetylene welding	Oxygen + acetylene flame (~3200 °C)	Mild steel filler rod (optional)	Melts steel ~1500 °C	Yes	Very strong	Repair work, thin steel, cutting (oxy-fuel cutting), pipework
MIG welding	Electric arc, continuous wire electrode	Wire electrode	Melts parent metal	Yes	Very strong	Car bodies (often robotic), steel fabrication, gates, furniture frames
Brazing	Gas torch/brazing hearth	Brass spelter + borax flux	~850–900 °C	No	Strong	Bike frames, mild steel fabrication, joining dissimilar metals
Hard (silver) soldering	Gas torch	Silver solder (Ag–Cu–Zn) + flux (borax/Easyflo)	~600–800 °C	No	Strong, neat	Jewellery, copper, brass, small steel parts, refrigeration pipes
Soft soldering	Soldering iron or blowtorch	Tin-based solder (lead-free: tin/copper/silver) + resin or active flux	~180–250 °C	No	Weak mechanically, good electrically	Electronics (PCBs), plumbing (copper pipes), tinplate

Key terms

Welding	Fusing metals by melting the edges of the parent metal (often with a filler rod) so they flow together and solidify as one piece. The strongest permanent joint.
Oxy-acetylene welding	Gas welding using oxygen and acetylene burnt at the torch nozzle (~3200 °C) to melt the joint, with a filler rod added by hand.
MIG welding	Metal Inert Gas: an electric arc between a continuously fed wire electrode and the work melts both, while a shielding gas (argon/CO_2) protects the weld pool.
Brazing	Joining with a brass (copper–zinc) filler (spelter) melting around 850–900 °C. The parent metals do not melt.
Hard soldering (silver soldering)	Joining with silver solder (silver–copper–zinc) melting about 600–800 °C – strong, neat joints in copper, brass, silver and steel.
Soft soldering	Joining with a low-melting tin-based solder (180–250 °C) – electronics and plumbing; weak mechanically.
Flux	Chemical that cleans and prevents oxidation of the joint during heating, helping filler to flow (borax for brazing/hard solder, resin for electronics).
Capillary action	The natural flow of molten filler into a narrow gap, even against gravity.
Heat-affected zone	Area around a weld where heating has altered the metal's properties.

Exam tip: Always state **"the parent metal does not melt"** when describing brazing/soldering, and link the choice to the product (e.g. soft solder for a PCB because higher temperatures would damage components).

3.4d Wood joints and knock-down fittings

Spec: Jointing – traditional wood joints, knock-down fittings (including use of specialist tools)

Key facts

• **Name five traditional wood joints.** Mortise and tenon, dovetail, housing, halving, dowel, finger/comb, mitre, lap/rebate, bridle, butt.
• **Which joint suits a table leg to rail?** Mortise and tenon (or dowel/bridle).
• **Which joint suits a drawer front?** Lap (stopped) dovetail – strong against the pulling force and attractive.
• **Which joint for shelves in a bookcase?** Housing joint (through or stopped housing).
• **Name four knock-down fittings.** Cam lock (cam and dowel), barrel nut and bolt (cross dowel), corner/modesty blocks, scan fittings/connector bolts, panel connectors, threaded inserts.
• **Advantages of KD fittings?** Flat-pack – cheaper storage and transport, home assembly with simple tools, can be dismantled and moved, no specialist skill.
• **Disadvantages of KD fittings?** Less strong/rigid than glued joints, can work loose, visible fittings, can damage chipboard when reassembled.
• **Tools for cutting a mortise and tenon?** Mortise gauge, try square, marking knife, tenon saw, mortise chisel and mallet (or mortiser machine), shoulder plane.
• **What is a dowelling jig?** A guide that clamps on the work so dowel holes are drilled accurately and in line in both pieces.

Joint	Description	Strength / features	Uses
Butt	Ends simply glued/nailed together	Weak – small gluing area	Simple boxes, with reinforcement
Dowel	Wooden pins in drilled holes	Strong, quick with a jig, invisible; can be used with manufactured boards	Frames, flat-pack furniture, joining boards
Mortise and tenon	Tenon on rail fits mortise in leg/stile	Very strong frame joint, large gluing area; through or stopped (blind)	Tables, chairs, doors, the rocking chair in 2019 Q4a
Bridle	Open slot over a tongue	Strong, good for frames	Frames, table legs/rails
Halving	Half thickness removed from both parts (corner, T, cross)	Moderate strength, flush faces	Frames, crossing rails
Housing	Groove across the grain; through or stopped	Supports shelves well	Bookcases, cabinets
Dovetail	Interlocking pins and tails (through, lap/stopped)	Very strong in one direction, attractive, skilled	Drawers, quality boxes
Finger (comb)	Interlocking rectangular fingers	Large gluing area, decorative	Boxes, machine-made joints (CNC router)
Mitre	Both pieces cut at 45°	Neat (no end grain) but weak – reinforce with splines/keys	Picture frames, mouldings
Lap / rebate	One piece rebated to take the other	Easy, hides end grain partially	Box corners, drawers (cheap)
Biscuit	Oval beech biscuit in slots	Aligns and strengthens panels	Worktops, cabinet carcasses

Fitting	How it works	Uses
Cam lock (cam and dowel)	A threaded cam dowel screws into one panel; the cam in a drilled recess in the other is rotated to grip and pull it tight	Wardrobes, bookcases, chests of drawers
Barrel nut and bolt (cross dowel)	A bolt passes through one part into a threaded barrel set crossways in the other	Bed frames, desks, chairs
Corner / modesty blocks	Plastic blocks screwed into both panels	Joining carcasses at 90°, fixing backs
Scan fittings / connector bolts	Bolt with flat head and threaded insert or sleeve	Table legs, beds
Panel connectors / threaded inserts	Threaded insert screwed into chipboard; machine screw joins parts	Repeated assembly in particle boards

Key terms

Mortise and tenon	Strong frame joint: a tenon (tongue) on one rail fits into a rectangular mortise (hole) in another. Used in tables, chairs and doors.
Dovetail joint	Interlocking fan-shaped tails and pins that resist pulling apart in one direction – drawers, quality boxes.
Housing joint	A groove (housing) cut across the grain to take the end of another board – shelves.
Halving joint	Half the thickness is removed from each piece so they overlap flush – frames, crossing rails.
Dowel joint	Wooden pins glued into matching holes – quick, accurate with a dowelling jig, used in flat-pack and frames.
Knock-down (KD) fitting	Mechanical fitting that allows furniture to be assembled and dismantled with simple tools – cam locks, barrel nuts, corner blocks.
Cam lock	KD fitting: a cam (disc) in one panel rotates to pull a dowel/bolt screwed into the other panel tight.
Biscuit joint	Oval compressed beech 'biscuit' glued into slots cut by a biscuit jointer – aligns and strengthens board joints.

Exam tip (2023 Q1a): "Screws" and "nuts and bolts" alone were **not** accepted as knock-down fittings – name them precisely (cam lock, barrel nut).

3.5a Finishes and preventing degradation

Spec: Finishes – paints, varnishes, sealants, preservatives, anodising, electro-plating, powder coating, oil coating, galvanisation, cathodic protection (including use of specialist tools)

Key facts

• **Why apply a finish?** Protect against corrosion/rot/UV/wear and moisture, improve aesthetics, make hygienic/easy to clean, add texture or grip, give information/branding.
• **What is anodising and which metal?** Electrolytic thickening of the oxide layer on aluminium; can be dyed; hard, corrosion resistant, used on bike parts, phone cases, carabiners.
• **Electroplating process?** Product is the cathode, plating metal is the anode, both in an electrolyte; DC current transfers metal ions onto the product.
• **Powder coating benefits?** Tough, even, thick finish with no runs; no solvents (low VOCs); overspray can be recycled; many

colours – bike frames, garden furniture, appliances.
• **Galvanising – why is it so good?** Zinc coating is tough and protects sacrificially even if scratched – motorway barriers, nails, buckets, steel beams, fencing.
• **Explain cathodic protection.** A more reactive metal (zinc/magnesium anode) is electrically connected to steel and corrodes instead of it – ship hulls, pipelines, oil rigs.
• **Varnish vs paint?** Varnish is a transparent protective coating that shows the grain; paint is opaque and adds colour (primer → undercoat → topcoat).
• **Benefits of varnish on a teak boat deck?** Seals against sea water/weathering, shows natural beauty of grain, protects from wear, smooth splinter-free and easy to clean.
• **What is oil coating?** Rubbing oil into wood (teak, Danish, linseed oil) or applying oil to steel to repel moisture – natural look, easy to reapply.
• **Name a timber preservative method.** Pressure treatment (tanalising) forcing preservative deep into the wood – fence posts, decking, sheds.

Finish	Materials	Process	Pros / cons	Examples
Paints	Wood, metal, some polymers	Prepare (sand/degrease) → primer → undercoat → top coats; brush, roller, spray (spray booth)	Colour, protection; can chip, needs maintenance; solvent paints release VOCs	Doors, cars, toys (non-toxic)
Varnishes	Wood (also paper/board, see 3.5b)	Brush/spray thin coats, sand lightly between (polyurethane, yacht varnish)	Clear, tough, waterproof, shows grain; can yellow/peel outdoors	Floors, furniture, boat decks
Sealants	Wood, joints, porous materials	Sanding sealer before finishing; silicone/acrylic sealants in gaps	Stop moisture/air entering; prepare surface	Baths, windows, sealing MDF edges
Preservatives	Timber	Brush, dip or **pressure treatment** (tanalising) with fungicide/insecticide	Long-life protection against rot and insects; toxic chemicals	Fences, sheds, decking, outdoor play equipment
Anodising	**Aluminium**	Clean/etch → aluminium is the **anode** in a sulfuric acid bath; current thickens the oxide layer → dye → seal in hot water	Hard, corrosion and wear resistant, bright colours, does not peel	Bike parts, torches, phone cases, carabiners
Electroplating	Metals (and plated plastics)	Product = **cathode**, plating metal = anode, in an electrolyte; DC current deposits metal	Decorative and protective thin layer (chrome, nickel, zinc, gold, silver); toxic chemicals	Taps, jewellery, cutlery, car trim
Powder coating	Metals (mostly steel, aluminium)	Degrease/pre-treat → electrostatically charged powder sprayed onto earthed part → cured in oven ~200 °C	Thick, tough, even, no runs, no solvents, overspray reused; needs oven, large items limited by oven size	Bike frames, garden furniture, radiators, the spin bike frame (2024 Q11)
Oil coating	Wood; steel tools	Rub in teak/Danish/linseed oil; light machine oil on steel	Natural look, enhances grain, easy to reapply; needs regular maintenance	Kitchen boards, teak furniture, tools
Galvanising	Steel/iron	Clean (degrease, pickle in acid, flux) → **hot-dip in molten zinc (~450 °C)** → cool	Very durable, protects **sacrificially** even if scratched; dull grey finish, can't be painted easily without preparation	Crash barriers, nails, fences, lamp posts, buckets
Cathodic protection	Steel structures	Attach **sacrificial anodes** (zinc, magnesium, aluminium) or apply an impressed current	Protects large structures in water/soil; anodes must be replaced	Ship hulls, pipelines, oil rigs, water heaters

Key terms

Corrosion	Chemical deterioration of a metal, e.g. rusting of iron/steel when exposed to oxygen and water (faster with salt).
Degradation	Breakdown of a material's properties over time – rot and insect attack in wood, UV embrittlement of polymers, corrosion of metals.
Anodising	Electrolytic process that thickens the natural oxide layer on aluminium; the porous layer can be dyed bright colours and is then sealed.
Electroplating	Depositing a thin layer of one metal (chromium, nickel, zinc, silver, gold) onto another using electrolysis – for appearance and corrosion protection.
Powder coating	Electrostatically charged dry polymer powder is sprayed onto earthed metal then cured in an oven to form a tough, even plastic coating.
Galvanising	Coating steel with zinc, usually by hot-dip immersion in molten zinc at ~450 °C. Zinc protects sacrificially.
Cathodic protection	Protecting steel by connecting it to a more reactive 'sacrificial anode' (zinc, magnesium, aluminium) that corrodes instead – or by an impressed electric current.
Preservative	Chemical treatment (often pressure impregnated) that protects timber from rot, fungi and insects.
Sealant	Liquid or flexible compound that fills gaps/seals surfaces against water and air (e.g. silicone, sanding sealer for wood).

3.5b Paper and board finishing

Spec: Paper and board finishing process – laminating, varnishing, hot foil blocking, embossing (including use of specialist tools)

Key facts

• **Benefits of laminating packaging?** Protects against moisture, dirt, scuffs and tearing; adds gloss or matt/soft-touch premium feel; makes colours richer.
• **Drawback of laminating?** Plastic film makes recycling more difficult; adds cost; can peel/bubble.
• **What is spot UV varnishing used for?** To highlight logos/text with a glossy contrast against a matt background and add protection.
• **How is hot foil blocking done?** A heated (≈100–200 °C) engraved metal die presses foil onto the board; the foil sticks only where the die touches.
• **Embossing vs debossing?** Embossing raises the image above the surface; debossing presses it into the surface. Both use matched male/female dies.
• **Why use these finishes on luxury packaging?** They add perceived quality, tactile appeal and shelf impact, helping to attract customers and justify higher prices.
• **Specialist equipment?** Laminating machines (rollers, heat), UV coaters and curing lamps, hot foil blocking press with heated dies, embossing press with male/female dies, die-cutters.

Finish	Process	Benefits	Drawbacks	Examples
Laminating	Thin polymer film (gloss, matt, silk, soft-touch) bonded to the sheet with heat and pressure, or adhesive, via rollers	Protects against moisture, dirt, tearing and scuffing; richer colours; premium feel	Harder to recycle (plastic + paper), extra cost, can curl/bubble	Book covers, menus, ID cards, luxury packaging

Finish	Process	Benefits	Drawbacks	Examples
Varnishing	Clear varnish applied on press (overall) or as **spot UV varnish**, cured by UV light	Gloss or matt effect, highlights areas, some protection, quick	Less protection than laminating; spot varnish needs accurate registration	Business cards, magazine covers, packaging highlights
Hot foil blocking	Heated engraved metal die presses metallic/coloured/holographic **foil** onto the surface	Striking metallic effect, luxury look, security features	Needs a custom die (cost), limited fine detail on textured stock	Perfume boxes, chocolate packaging, wedding invitations, book spines
Embossing / debossing	Sheet pressed between matched **male and female dies** to raise (emboss) or sink (deboss) an image; can combine with foil or print (registered), or be **blind**	Tactile, 3D effect, premium feel, no ink needed	Die cost; can weaken thin board; image visible on reverse	Logos on packaging, greetings cards, letterheads, certificates

Key terms

Laminating (paper)	Bonding a thin plastic film (gloss, matt, soft-touch) to paper or board using heat and pressure or adhesive.
Varnishing (print)	Applying a clear coating to printed work – overall or as a spot UV varnish on selected areas for contrast and protection.
Spot UV varnish	Varnish printed only on chosen areas and cured instantly with ultraviolet light, giving a glossy highlight.
Hot foil blocking	A heated metal die presses metallic or coloured foil onto the surface so it transfers in the shape of the die.
Embossing	Pressing paper/board between male and female dies to create a raised image. Debossing creates a recessed image.
Blind embossing	Embossing without ink or foil – the raised image alone.

Exam tip: Name the process precisely and link it to the target market – e.g. "hot foil blocking adds a metallic logo *so* the packaging looks premium *which* attracts customers and justifies a higher price."